Probability and Statistics

Discrete Random Variables and Probability Distributions

(PART 2)

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1 Geometric Distribution

2 Hypergeometric and Negative Binomial Distributions

3 The Poisson Probability Distribution

Geometric Distribution

- Geometric random variable (几何随机变量)
- An experiment consists of a sequence of trials
- Each trial has two possible outcomes
  - For convenience, denote the two outcomes as S and F
- The trials are independent
- The probability of S is fixed, denoted as p
- Geometric random variable X is the number of trials until the first S occurs
- Its PMF is $p_X(x) = p(1-p)^{x-1}$

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- What is CDF of X?

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Geometric Distribution

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- What is CDF of X ?

$$P[X > n] = \sum_{x=n+1}^{\infty} p(1-p)^{x-1}$$

$$= p \sum_{x=n+1}^{\infty} (1-p)^{x-1} = (1-p)^n p \sum_{x=n+1}^{\infty} (1-p)^{x-n-1}$$

$$= (1-p)^n p \sum_{x=0}^{\infty} (1-p)^x = \frac{(1-p)^n p}{p}$$

$$= (1-p)^n$$

$$F_X(x) = P[X \le x] = \sum_{k=1}^{x} p(1-p)^{k-1} = 1 - P[X > x] = 1 - (1-p)^x$$

Geometric Distribution

- Expected value (期望值)
- A worker continuously checks products coming from a product line until a defective one is found, and assume P(D) = p
- N = the number of products that are checked
- PMF of N is

$$p_N(n) = p(1-p)^{n-1} \qquad n = 1, 2, \ldots$$

- What is E[N]?

$$E[N] = \sum_n np_N(n) = p\sum_{n=1}^{\infty} n(1-p)^{n-1}$$

$$\sum_{n=1}^{\infty}(1-p)^n = \frac{1}{1-(1-p)} - 1 = \frac{1}{p} - 1$$

$$E[N] = -p\frac{d}{dp}\sum_{n=1}^{\infty}(1-p)^n = -p\frac{d}{dp}\left\{\frac{1}{p} - 1\right\} = -p\left\{-\frac{1}{p^2}\right\} = \boxed{\frac{1}{p}}$$

$$\frac{d}{dp}\sum_{n=1}^{\infty}(1-p)^n = \sum_{n=1}^{\infty}\frac{d}{dp}(1-p)^n = -\sum_{n=1}^{\infty}n(1-p)^{n-1} = -p^{-2}$$

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Geometric Distribution

- Variance (方差)
- A worker continuously checks products coming from a product line until a defective one is found, and assume P(D) = p
- N = the number of products that are checked
- PMF of N is

$$p_N(n) = p(1-p)^{n-1} \qquad n = 1, 2, \ldots$$

- What is V[N]?

$$V[N] = E[N^2] - E[N]^2 = \frac{2-p}{p^2} - (\frac{1}{p})^2 = \boxed{\frac{1-p}{p^2}}$$

$$\frac{d^2}{dp^2}\sum_{n=1}^{\infty}(1-p)^n = \sum_{n=1}^{\infty} n(n-1)(1-p)^{n-2} = \frac{1}{1-p}\sum_{n=1}^{\infty} n(n-1)(1-p)^{n-1} = 2p^{-3}$$

$$\sum_{n=1}^{\infty}(1-p)^n = \frac{1}{p} - 1$$

$$\frac{d^2}{dp^2}\sum_{n=1}^{\infty}(1-p)^n = 2p^{-3}$$

$$\sum_{n=1}^{\infty} n(n-1)(1-p)^{n-1} = 2p^{-3}(1-p)$$

$$\sum_{n=1}^{\infty} n^2(1-p)^{n-1} = 2p^{-3}(1-p) + \sum_{n=1}^{\infty} n(1-p)^{n-1} = 2p^{-3}(1-p) + p^{-2}$$

Geometric Distribution

- Geometric random variable (几何随机变量)
- A couple is planning to have children where the probability of having a boy is p.
- If they keep trying until they have a girl and then stop, what is the expected number of children that they will have?

Geometric Distribution

- Geometric random variable (几何随机变量)
- A couple is planning to have children where the probability of having a boy is p.
- If they keep trying until they have a girl and then stop, what is the expected number of children that they will have?
- Let X = number of childeren they will have, then X is a geometric random variable, hence $E[X] = \frac{1}{1-p}$

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- Let X = number of childeren they will have, then X is a geometric random variable, hence $E[X] = \frac{1}{1-p}$
- Suppose they plan to have a minimum of four children. However, if the first four are all boys, they will continue until they have a girl and then stop. If there is at least one girl among the first four children, they will stop at four. What is the expected number of children that they will have?

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$E[X]$

$= P(at\ least\ one\ girl\ in\ first\ four\ children) * (the\ number\ of\ first\ four\ children)$

$+P(first\ four\ children\ are\ all\ boys) * (the\ number\ of\ first\ four\ boys\ + additional\ number\ of\ children\ until\ a\ girl)$

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$= (1-p^4) * 4 + p^4 * (4 + \frac{1}{1-p})$

Geometric Distribution

- Geometric random variable (几何随机变量)
- We are given a box containing R red balls and B blue balls in which a ball is randomly selected from the box with replacement until a blue ball is selected. Let N be a random variable that denotes the number of trials until a blue ball is selected.
- What is PMF of N?

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$$p_N(n) = p(1-p)^{n-1} = \left(\frac{B}{R+B}\right)\left(\frac{R}{R+B}\right)^{n-1} \qquad n = 1, 2, \ldots$$

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- The probability that the experiment requires at least k trials before it stops is?

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- The probability that the experiment requires at least *k* trials before it stops is?

$$P[N \geq k] = 1 - P[N < k] = 1 - \sum_{n=1}^{k-1} p(1-p)^{n-1} = 1 - \frac{p[1-(1-p)^{k-1}]}{1-(1-p)} = (1-p)^{k-1} = \left(\frac{R}{R+B}\right)^{k-1}$$

Geometric Distribution

- Geometric random variable (几何随机变量)
- Let K denote the number of tries until we find a person who wears glasses in a city. The probability that a person wears a glass in this city is p = 0.2

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- Let K denote the number of tries until we find a person who wears glasses in a city. The probability that a person wears a glass in this city is $p = 0.2$
- The probability that it takes exactly 10 tries to get a person who wears glasses is?

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$$P[K = 10] = 0.2(0.8)^{10-1} = 0.2(0.8)^{9} = 0.02684$$

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$$P[K = 10] = 0.2(0.8)^{10-1} = 0.2(0.8)^{9} = 0.02684$$

- The probability that it takes at least 10 tries to get a person who wears glasses is?

$$P[K \geq 10] = 1 - P[K < 10] = 1 - \sum_{k=1}^{9} p(1-p)^{k-1} = (1-p)^{9} = (0.8)^{9} = 0.1342$$

Geometric Distribution

- Forgetfulness property of the geometric distribution（无记忆性性质）
- Let X be a random variable that denotes the number of trials until the first success. Suppose we have observed a number n of these trials, and they are all failures. We would like to know the number K of additional trials until the first success.
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$p_K(k) = P(K = k|X > n)$

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$$p_K(k) = P(K = k|X > n) = P(X = n + k|X > n)$$

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$$p_K(k) = P(K = k|X > n) = P(X = n + k|X > n)$$

$$= \frac{P(X = n + k \cap X > n)}{P(X > n)} = \frac{P(X = n + k)}{P(X > n)} = \frac{p(1-p)^{n+k-1}}{(1-p)^n}$$

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$$= p(1-p)^{k-1}$$

- So, we can find that the probability of number of required additional trials is the probability of required numbers counting from the start

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Geometric Distribution

- Forgetfulness property of the geometric distribution（无记忆性性质）
- The result shows that the conditional probability that the number of trials remaining until the first success, given that no success occurred in the first n trials, has the same PMF as X had originally
- It means that the distribution "forgets" the past by not remembering how long the sequence has lasted if no success has already occurred
- Thus, each time we want to know the number of trials until the first success, given that the first success has not yet occurred, the process "starts" from the beginning

Geometric Distribution

- Forgetfulness property of the geometric distribution（无记忆性性质）
- There are only two probability distributions with forgetfulness proproperty
  - Geometric distribution
  - Exponential distribution
- Forgerfulness property is very important in the stochastic process of engineering
  - Markov property
    - Future state of the process depends only on its present state, not on the sequence of events that preceded it. In simpler terms, given the present, the future is independent of the past

$$P(X_{t+1} = x_{t+1} | X_t = x_t, X_{t-1} = x_{t-1}, \ldots, X_0 = x_0) = P(X_{t+1} = x_{t+1} | X_t = x_t)$$

    - Markov chain, Markov decision process (MDP), reinforcement learning ...

Geometric Distribution

- Forgetfulness property of the geometric distribution（无记忆性性质）
- One is tossing balls randomly into 50 boxes, and his goal is to stop when he gets the first ball into the eighth box. Given that he has tossed 20 balls without getting a ball into the eighth box, what is the expected number of additional tosses he needs to get a ball into the eighth box?

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- Let X be the number of tosses, then X is a geometric random variable. Let K be the number of additional tosses required to get a ball into the eighth box, then due to the forgetfulness property, $E[K] = 1 / p = 50$

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2 Hypergeometric and Negative Binomial Distributions

3 The Poisson Probability Distribution

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Hypergeometric and Negative Binomial Distributions

- Hypergeometric random variable（超几何随机变量）
- The entire number of individuals is N
- Each individual has two possible outcomes
  - For convenience, we denote the two oucomes as S and F
- The number of S's is M
- n individuals is selected from the entire N individuals
- Hypergeometric random variable X is the number of S`s in the selected n individuals
- Its PMF is denoted by $h(x; n, M, N) = \frac{C_{x,M} * C_{n-x,N-M}}{C_{n,N}}$

Hypergeometric and Negative Binomial Distributions

- Hypergeometric random variable（超几何随机变量）
  - There are N products, D of them are defective
  - What is the probability that k defective products are selected if m products are randomly selected (so the number of defective ones is k, and the number of intact ones is m - k)?
  - $$\frac{Number\ of\ ways\ that\ k\ defective\ ones\ selected * Number\ of\ ways\ m-k\ intact\ ones\ selected}{Number\ of\ ways\ m\ products\ are\ selected}$$
    - The number of combinations of picking m products is $C_{m,N}$
    - The number of combinations of picking k defective products is $C_{k,D}$
    - The number intact products in m selected products is m - k, the number of intact products in the total products is N - D
    - Hence, the number of ways to select m - k intact products from the total N - D intact products is $C_{m-k,N-D}$
    - Hence, $\frac{C_{k,D} * C_{m-k,N-D}}{C_{m,N}}$

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Hypergeometric and Negative Binomial Distributions

- The range of Hypergeometric random variable X
- $p_X(x) = h(x; n, M, N) = \frac{C_{x,M} * C_{n-x,N-M}}{C_{n,N}}$
- If $n \leq M$,

Hypergeometric and Negative Binomial Distributions

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- If n ≤ M, then X ≤ n

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- If N - M ≥ n,

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- If n ≤ M, then X ≤ n
- If n > M, then X ≤ M
- If N - M ≥ n, then X ≥ 0

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- If N - M ≥ n, then X ≥ 0
- If N - M < n,

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- If n ≤ M, then X ≤ n
- If n > M, then X ≤ M
- If N - M ≥ n, then X ≥ 0
- If N - M < n, then X ≥ n - (N - M)

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Hypergeometric and Negative Binomial Distributions

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- So, the range of X is

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- If n ≤ M, then X ≤ n
- If n > M, then X ≤ M
- If N - M ≥ n, then X ≥ 0
- If N - M < n, then X ≥ n - (N - M)
- So, the range of X is
- max(n - (N - M), 0) ≤ X ≤ min(n, M)

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Hypergeometric and Negative Binomial Distributions

- A student knows the answers of 8 out of 12 questions. Now the final term exam contains 6 questions.
- The probability that the student is able to solve 4 or more problems?

Hypergeometric and Negative Binomial Distributions

- A student knows the answers of 8 out of 12 questions. Now the final term exam contains 6 questions.
- The probability that the student is able to solve 4 or more problems?
- Let K be a random variable that denotes the number of questions that the student can solve. Then its PMF is

$$p_K(k) = h(k; 6,8,12) = \frac{C_{k,8} * C_{6-k,4}}{C_{6,12}}$$

$$P[K \geq 4] = \sum_{k=4}^{6} \left\{ \frac{\binom{8}{k}\binom{4}{6-k}}{\binom{12}{6}} \right\} = \frac{\binom{8}{4}\binom{4}{2} + \binom{8}{5}\binom{4}{1} + \binom{8}{6}\binom{4}{0}}{\binom{12}{6}} = \frac{420 + 224 + 28}{924} = \frac{672}{924}$$

$$= 0.7273$$

Hypergeometric and Negative Binomial Distributions

- A class has 18 boys and 12 girls. If the teacher randomly selects a group of 15 students, determine the following:
- The probability that 8 members of the group are girls?

Hypergeometric and Negative Binomial Distributions

- A class has 18 boys and 12 girls. If the teacher randomly selects a group of 15 students, determine the following:
- The probability that 8 members of the group are girls?
- Let *N* be a random variable that denotes the number of girls among a group of 15

$$p_N(n) = h(n; 15,12,30) = \frac{C_{n,12} \ * \ C_{15-n,18}}{C_{15,30}}$$

$$P[N = 8] = \frac{\binom{12}{8}\binom{18}{7}}{\binom{30}{15}} = 0.10155$$

Hypergeometric and Negative Binomial Distributions

- Negative binomial random variable（负二项随机变量）
- An experiment consists of a sequence of trials
- Each trial has two possible outcomes
  - For convenience, denote the two outcomes as S and F
- The trials are independent
- The probability of S is fixed, denoted as p
- The experiment continues until the number of S's is *r*
- Negative binomial random variable X is the number of Fs
- Its PMF is denoted by *nb(x; r, p)*
- In contrast to the binomial random variable, in negative binomial random variable the number of S's is fixed, but the number of trials is not fixed

Hypergeometric and Negative Binomial Distributions

- Negative binomial random variable（负二项随机变量）
- The event {X = x} is equivalent to {r - 1 S's in the first x + r - 1 trials and an S on the (x + r) th trial}

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- r = 2 and x = 2, then there must be 1 S's in the first 3 trials and trial 4 must be an S

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- $nb(x; r, p)=C_{r-1,x+r-1} * p^{r-1} * (1-p)^x * p = C_{r-1,x+r-1} * p^r * (1-p)^x$

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- r = 2 and x = 2, then there must be 1 S's in the first 3 trials and trial 4 must be an S
- $nb(x; r, p) = C_{r-1,x+r-1} * p^{r-1} * (1-p)^x * p = C_{r-1,x+r-1} * p^r * (1-p)^x$

Binomial random variable
b(r-1;x+r-1,p)

Hypergeometric and Negative Binomial Distributions

- Twenty percent of the population of a particular city wear glasses. If you randomly stop people from that city, determine the following probabilities:
- It takes exactly 10 tries to get the third person who wears glasses

Hypergeometric and Negative Binomial Distributions

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- It takes exactly 10 tries to get the third person who wears glasses?
- Let K_r be a random variable that denotes the number of tries up to and including the try that results in the r th person who wears glasses. Since the probability of success on any try is $p = 0.2$, the PMF of K_r is given by

$$p_{K_r}(k) = \binom{k-1}{r-1}(0.2)^r(0.8)^{k-r} \qquad k = r, r+1, \ldots; r = 1, 2, \ldots, k$$

Hence, the probability that it takes exactly 10 tries to get the 3 rd person who wears glasses is given by

$$P[K_3 = 10] = p_{K_3}(10) = \binom{10-1}{3-1}(0.2)^3(0.8)^7 = \binom{9}{2}(0.2)^3(0.8)^7 = 36(0.2)^3(0.8)^7 = 0.0604$$

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- It takes at least 10 tries to get the third person who wears glasses?

$$P[K_3 \geq 10] = 1 - P[K_3 < 10] = 1 - \sum_{k=3}^{9}\binom{k-1}{3-1}(0.2)^3(0.8)^{k-3} = 1 - \sum_{k=3}^{9}\binom{k-1}{2}(0.2)^3(0.8)^{k-3}$$

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Hypergeometric and Negative Binomial Distributions

- The probability of getting a head in a single toss of a biased coin is q.
- In an experiment that consists of repeated tosses of the coin, what is the probability that the 18 th head occurs on the 30 th toss?

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- Let N_k be a random variable that denotes the number of tosses up to and including the toss that results in the k th head. Then, its PMF is

$$p_{N_k}(n) = \binom{n-1}{k-1} q^k (1-q)^{n-k} \qquad n = k, k+1, \ldots; k = 1, 2, \ldots, n$$

Hence, the probability that the 18 th head occurs on the 30 th toss is given by

$$P[N_{18} = 30] = p_{N_{18}}(30) = \binom{30-1}{18-1} q^{18} (1-q)^{12} = \binom{29}{17} q^{18} (1-q)^{12} = 51895935 q^{18} (1-q)^{12}$$

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1 Geometric Distribution

2 Hypergeometric and Negative Binomial Distributions

3 The Poisson Probability Distribution

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The Poisson Probability Distribution

- Poisson random variable（泊松随机变量）
- Its PMF is denoted by $Poi(x;\lambda) = e^{-\lambda} * \frac{\lambda^x}{x!}$
- The Poisson distribution has many applications in science and engineering
  - The number of telephone calls arriving at a switchboard during an interval
    - λ usually stands for the average number of telephone calls arriving during an interval
    - For example, if the average number of telephone calls arriving during one hour is 20, the probability that 25 calls will arrive during one hour is

      $p(25;20) = e^{-20} * \frac{20^{25}}{25!} = 0.04$

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The Poisson Probability Distribution

- Poisson random variable（泊松随机变量）
- Messages arrive at a switchboard in a Poisson manner at an average rate of 6 per hour. Let K be the random variable that denotes the number of messages arriving at the switchboard within a one-hour interval
- What is PMF of K?

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- What is PMF of K?

$$p_K(k) = \left(\frac{6^k}{k!}\right)e^{-6} \quad k = 0, 1, 2, \ldots$$

The Poisson Probability Distribution

- Mean and variance of poisson distribution
- $E[X] = \lambda$, $Var[X] = \lambda$

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- $$E[X] = \sum_{k=0}^{\infty} \frac{k\lambda^k}{k!} e^{-\lambda}$$

$$= \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

$$= \lambda e^{-\lambda} e^{\lambda}$$

$$= \lambda$$

The Poisson Probability Distribution

- Mean and variance of poisson distribution
- E[X] = λ, Var[X] = λ

- $E[X^2] = \sum_{k=0}^{\infty} k^2 p_K(k) = \sum_{k=0}^{\infty} \frac{k^2 \lambda^k}{k!} e^{-\lambda}$

$$= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{k\lambda^{k-1}}{(k-1)!}$$

But

$$\sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} = \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{\lambda}$$

$$\frac{d}{d\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} = \sum_{k=1}^{\infty} \frac{d}{d\lambda} \left[\frac{\lambda^k}{(k-1)!} \right] = \sum_{k=1}^{\infty} \frac{k\lambda^{k-1}}{(k-1)!} = \frac{d}{d\lambda} (\lambda e^{\lambda}) = e^{\lambda}(1+\lambda)$$

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But

$$= \lambda^2 + \lambda$$

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$$= \lambda^2 + \lambda$$

Var[X] = E[X^2] - $(E[X])^2 = \lambda$

But

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The Poisson Probability Distribution

- Poisson Approximation to the Binomial Distribution
- Let X be a binomial random variable $b(n,p)$ and PMF

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \ldots, n$$

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$e^{-\lambda}$

$$\lim_{n\to\infty}\left(1-\frac{a}{n}\right) = 1$$

$$\lim_{n\to\infty}\left(1+\frac{a}{n}\right)^n = e^a$$

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The Poisson Probability Distribution

- Poisson Approximation to the Binomial Distribution

$$b(x; n, p) = Poi(x; np)$$

$$C_{x,n} * p^x * (1-p)^{n-x} = e^{-np} * \frac{(np)^x}{x!}$$

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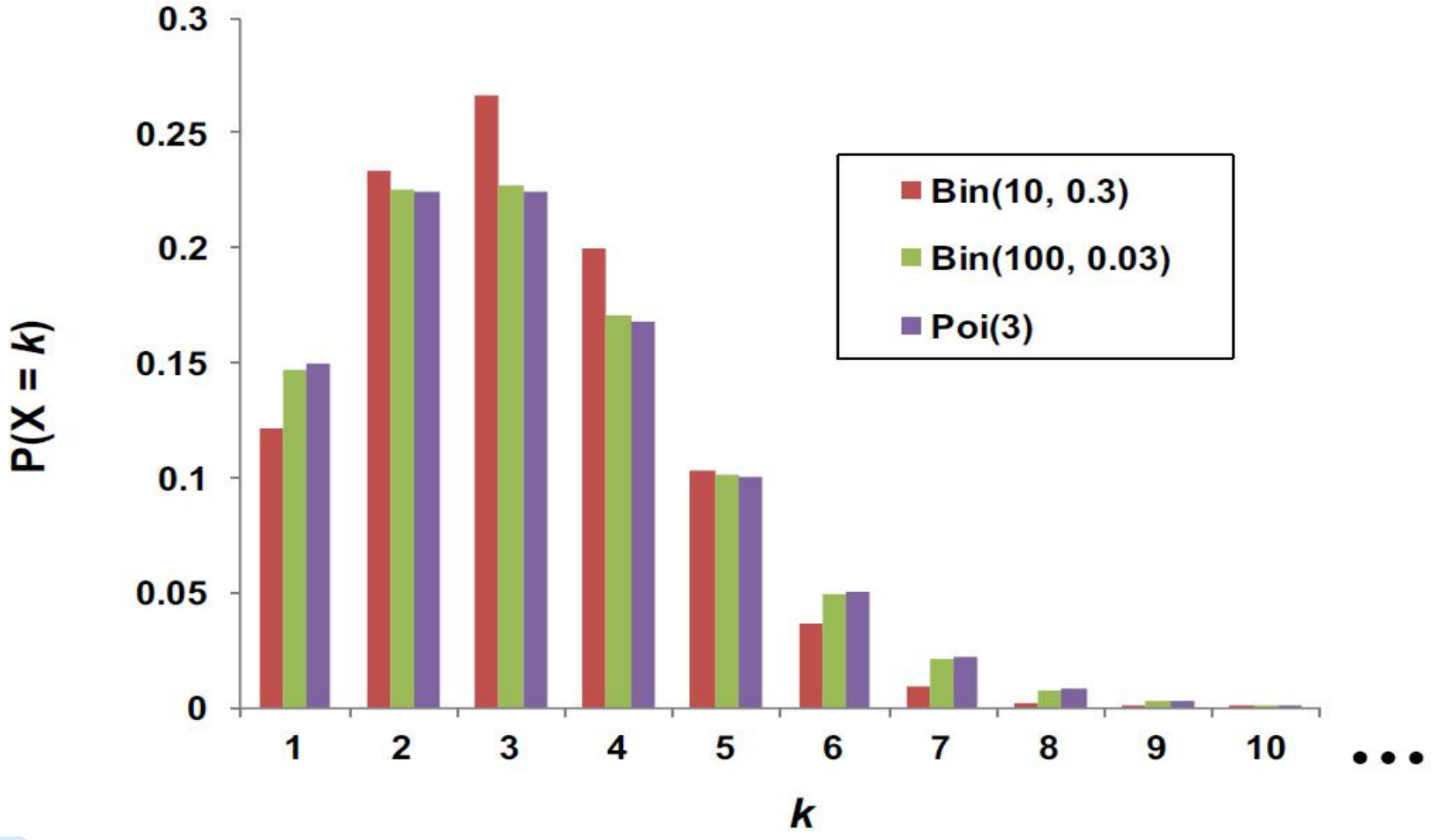
$$\lim_{n\to\infty}\left(1-\frac{a}{n}\right)=1$$

$$\lim_{n\to\infty}\left(1+\frac{a}{n}\right)^n=e^a$$

- So, this approximation is possible only when n is very large, p is very small

The Poisson Probability Distribution

- Poisson Approximation to the Binomial Distribution
- *b(10, 0.3) vs b(100, 0.03) vs Poi(3)*



The Poisson Probability Distribution

- Massive machine type communications of 6G (mMTC)
- There are two different types of node, A and B, the number of nodes A is n_1, B is n_2, transmission probability of A and B type are p_1 and p_2 respectively
- A successful transmission of type A and B node takes s_1 and s_2 seconds respectively, a failure transmission takes f seconds, and idle slot takes i seconds
- Throughput of wireless network?
- Because in 6G Network, the number of nodes is very huge, i.e., $n_1 = \infty, n_2 = \infty$, we can use the approximation method

- Throughput $= \frac{Probability\ of\ success}{Average\ length\ of\ slot}$
- $P(s_1) = C_{1,n1} * p_1 * (1-p_1)^{n1-1} * (1-p_2)^{n2} =$?
- $P(s_2) = C_{1,n2} * p_2 * (1-p_2)^{n2-1} * (1-p_1)^{n1} =$?
- P(s) = $P(s_1) + P(s_2)$
- P(i) = $(1-p_1)^{n1} * (1-p_2)^{n2} =$?
- P(f) = 1 - P(s) - P(i)
- Average length of slot =s_1 * P(s_1) + s_2 * P(s_2) + i * P(i) + f * P(f)
- Throughput $= \frac{P(s)}{Average\ length\ of\ slot}$

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- $$P(s_1) = C_{1,n1} * p_1 * (1-p_1)^{n1-1} * (1-p_2)^{n2} = e^{-n_1p_1} * \frac{(n_1p_1)^{1}}{1!} * e^{-n_2p_2} * \frac{(n_1p_1)^{0}}{0!}$$
$$= e^{-n_1p_1} * n_1p_1 * e^{-n_2p_2}$$

- $$P(s_2) = C_{1,n2} * p_2 * (1-p_2)^{n2-1} * (1-p_1)^{n1} = e^{-n_2p_2} * \frac{(n_2p_2)^{1}}{1!} * e^{-n_1p_1} * \frac{(n_1p_1)^{0}}{0!}$$
$$= e^{-n_2p_2} * n_2p_2 * e^{-n_1p_1}$$

- $$\mathrm{P(i)} = (1-p_1)^{n1} * (1-p_2)^{n2} = e^{-n_1p_1} * \frac{(n_1p_1)^{0}}{0!} * e^{-n_2p_2} * \frac{(n_1p_1)^{0}}{0!} = e^{-n_1p_1} * e^{-n_2p_2}$$

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- Poisson process (泊松过程）
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- Suppose the number of cars arriving at a parking lot follows a Poisson process at an average rate of 6 per hour. What is the probability that there are more than one arriving cars during 30 minutes?

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- Suppose the number of cars arriving at a parking lot follows a Poisson process at an average rate of 6 per hour. What is the probability that there are more than one arriving cars during 30 minutes?
- $\lambda = 6 * 0.5 = 3$
- $P(1 \leq X) = 1 - P(X = 0) = 1 - e^{-3} * \frac{3^0}{0!} = 0.95$

The Poisson Probability Distribution

- The number of cars that arrive at a gas station is a Poisson random variable with a mean of 50 cars per hour. The station has only one attendant, and each car requires exactly one minute to fill up. If we define a waiting line as the condition in which two or more cars are found at the same time at the gas station.
- What is the probability that a waiting line will occur at the station?

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- What is the probability that a waiting line will occur at the station?
- Let N denote the number of cars that arrive at the gas station. Since N is a Poisson random variable with a mean of cars per minute, the PMF is given by

$$p_N(n) = \frac{(5/6)^n}{n!} e^{-(5/6)} \qquad n = 0, 1, 2, \ldots$$

Since it takes exactly 1 minute to service a car, a waiting line occurs when at least 1 other car arrives within the 1-minute interval it takes to finish serving the current car receiving service. Thus, the probability that a waiting line will occur at the station is given by

$$P[N > 0] = 1 - P[N = 0] = 1 - e^{-(5/6)} = 0.5654$$

The Poisson Probability Distribution

- The number of traffic tickets that a officer gives out has been shown to have a Poisson distribution with a mean of 7
- What is the probability that the officer gave out no ticket?

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- What is the probability that the officer gave out no ticket?
- Let K denote the number of traffic tickets that the officer gives out. Its PMF is

$$p_K(k) = \frac{7^k e^{-7}}{k!} \qquad k = 0, 1, 2, \ldots$$

The probability that on one particular day the officer gave out no ticket is given by

$$P[K = 0] = p_K(0) = e^{-7} = 0.0009$$

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- What is the probability that he gives out fewer than 4 tickets?

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- Let K denote the number of traffic tickets that the officer gives out. Its PMF is

$$p_K(k) = \frac{7^k e^{-7}}{k!} \qquad k = 0, 1, 2, \ldots$$

The probability that on one particular day the officer gave out no ticket is given by

$$P[K = 0] = p_K(0) = e^{-7} = 0.0009$$

- What is the probability that he gives out fewer than 4 tickets?

$$P[K < 4] = p_K(0) + p_K(1) + p_K(2) + p_K(3) = e^{-7}\left\{1 + 7 + \frac{49}{2} + \frac{343}{6}\right\} = 0.0818$$